Topic 11: Division by t

Unit IV — Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

Contents

1 Opening Hook

Hook

In Topic 10, you discovered that multiplying $f(t)$ by t differentiates $F(s)$. Now the beautiful mirror image: dividing $f(t)$ by t integrates $F(s)$ from s to ∞. This gives you the transform of functions like $\sin(t)/t$ — a function that appears in communication theory (the sinc function) and is famously difficult to integrate directly. With the division-by-t property, its Laplace transform is just

$$\int_s^\infty \frac{1}{u^2+1}\,du \;=\; \frac{\pi}{2} - \arctan(s).$$

Elegant, fast, and powerful.

2 Why This Topic Exists

"Before we begin, here is the honest answer to why you are reading this..." Functions divided by t appear throughout applied mathematics and engineering. The sinc function $\sin(t)/t$ is the impulse response of an ideal low-pass filter in signal processing. The ratio $(e^{at} - e^{bt})/t$ appears in heat-flow problems and integral evaluations. Without the division-by-t property we would be forced into difficult direct integrations; with it the answer is one clean integral of $F(s)$ in the s-domain.

Mistake	**Consequence**
Not checking the existence condition $\lim_{t\to 0^+} f(t)/t$	Applying the formula to a non-integrable f
Integrating from 0 to s instead of s to ∞	Completely wrong answer — direction of i
Confusing with the integral transform formula (Topic 09)	Using $F(s)/s$ instead of $\int_s^\infty F(u)\,du$

Key Warning

Always verify the existence condition *before* applying $\mathcal{L}\left\{\dfrac{f(t)}{t}\right\} = \int_s^\infty F(u)\,du$. The integration runs from s up to ∞, not the other way.

3 You Already Know This (Intuition First)

Recall from ordinary calculus that differentiation and integration are inverse operations. The same elegant duality appears in the Laplace world:

- **Multiply by** t in the time domain $\longleftrightarrow$ **differentiate** $F(s)$ (Topic 10).
- **Divide by** t in the time domain $\longleftrightarrow$ **integrate** $F(s)$ from s to ∞ (this topic).

Just as integration "un-differentiates" a function, dividing by t "un-differentiates" in the s-domain — it recovers the integral of $F(s)$ from s to ∞. This Topics 10–11 duality mirrors the derivative-integral duality you already know, making the new formula easy to remember: *multiply* $\to$ *differentiate; divide* $\to$ *integrate.*

4 The Main Theorem — Division by t

Theorem: Division by t

Theorem. If $\mathcal{L}\{f(t)\} = F(s)$ and $\lim_{t\to 0^+} \dfrac{f(t)}{t}$ exists (is finite), then

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u)\,du.$$

Proof. Start from the definition of $\int_s^\infty F(u)\,du$:

$$\int_s^\infty F(u)\,du = \int_s^\infty \left[\int_0^\infty e^{-ut} f(t)\,dt\right] du.$$

Both iterated integrals converge absolutely (since f is of exponential order and the existence condition guarantees no singularity at $t=0$), so Fubini's theorem permits interchange of order:

$$= \int_0^\infty f(t)\left[\int_s^\infty e^{-ut}\,du\right] dt.$$

Evaluate the inner integral: for $t > 0$ and $\mathrm{Re}(s) > 0$,

$$\int_s^\infty e^{-ut}\,du = \left[\frac{e^{-ut}}{-t}\right]_s^\infty = 0 - \frac{e^{-st}}{-t} = \frac{e^{-st}}{t}.$$

Substituting back:

$$\int_s^\infty F(u)\,du = \int_0^\infty f(t)\cdot\frac{e^{-st}}{t}\,dt = \int_0^\infty e^{-st}\frac{f(t)}{t}\,dt = \mathcal{L}\left\{\frac{f(t)}{t}\right\}. \qquad \square$$

Existence Condition. The theorem requires $\lim_{t\to 0^+} \dfrac{f(t)}{t}$ to be finite. If $f(0) \neq 0$, then $f(t)/t \sim f(0)/t \to \infty$ as $t \to 0^+$ and the Laplace integral diverges. Consequently we need $f(0) = 0$ (or more precisely, $\lim_{t\to 0^+} f(t)/t$ finite). For example, $\mathcal{L}\{\cos(t)/t\}$ does **not** exist because $\cos(0)/0 \to \infty$.

What Did We Learn?

$\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u)\,du$, proved by swapping integration order via Fubini. The existence condition $f(0) = 0$ is non-negotiable.

5 Key Application — $\mathcal{L}\{\sin(t)/t\}$

Derivation: $\mathcal{L}\{\sin(t)/t\}$

Let $f(t) = \sin t$. Then $\mathcal{L}\{\sin t\} = F(s) = \dfrac{1}{s^2+1}$.

Check existence condition: $\lim_{t\to 0^+} \dfrac{\sin t}{t} = 1$ (finite). ✓

Apply the theorem:

$$\mathcal{L}\left\{\frac{\sin t}{t}\right\} = \int_s^\infty \frac{du}{u^2+1} = \Big[\arctan(u)\Big]_s^\infty.$$

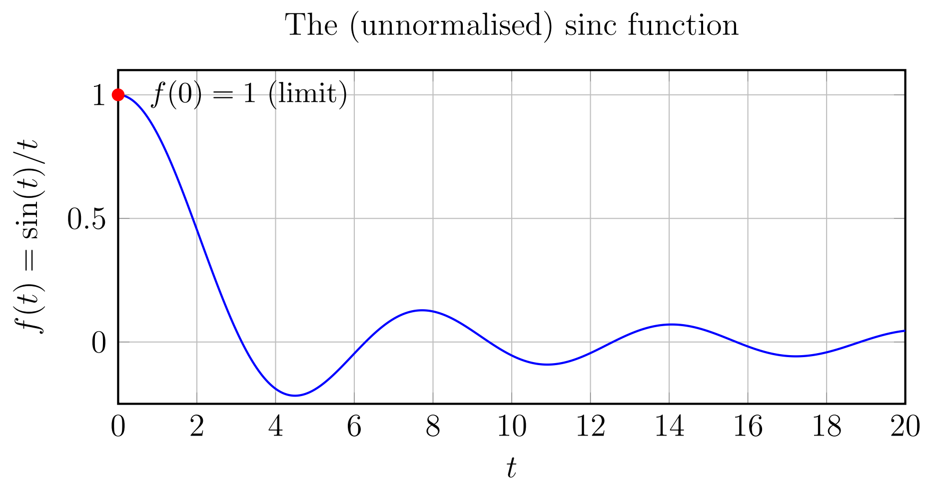
Now evaluate the limits:

$$\lim_{u\to\infty} \arctan(u) = \frac{\pi}{2}, \qquad \arctan(s) \text{ at the lower limit.}$$

Therefore:

$$\mathcal{L}\left\{\frac{\sin t}{t}\right\} = \frac{\pi}{2} - \arctan(s).$$

Graph of the sinc function $f(t) = \sin(t)/t$.



What Did We Learn?

$\mathcal{L}\left\{\dfrac{\sin t}{t}\right\} = \dfrac{\pi}{2} - \arctan(s)$. The key steps: (1) recognise $F(s) = 1/(s^2+1)$, (2) verify the existence condition, (3) integrate and evaluate using $\arctan(\infty) = \pi/2$.

6 Key Application — $\mathcal{L}\{(e^{at} - e^{bt})/t\}$

Derivation: $\mathcal{L}\{(e^{at} - e^{bt})/t\}$

Let $f(t) = e^{at} - e^{bt}$ (assume $s > \max(a, b)$ for convergence). Then

$$F(s) = \mathcal{L}\{e^{at}\} - \mathcal{L}\{e^{bt}\} = \frac{1}{s-a} - \frac{1}{s-b}.$$

Check existence condition:

$$\lim_{t\to 0^+} \frac{e^{at} - e^{bt}}{t} = \lim_{t\to 0} \frac{(1 + at + \cdots) - (1 + bt + \cdots)}{t} = a - b \quad \text{(finite)}. \quad \checkmark$$

Apply the theorem:

$$\mathcal{L}\left\{\frac{e^{at} - e^{bt}}{t}\right\} = \int_s^\infty \left[\frac{1}{u-a} - \frac{1}{u-b}\right] du = \Big[\ln(u-a) - \ln(u-b)\Big]_s^\infty = \left[\ln \frac{u-a}{u-b}\right]_s^\infty.$$

Evaluate the upper limit:

$$\lim_{u\to\infty} \ln \frac{u-a}{u-b} = \lim_{u\to\infty} \ln \frac{1 - a/u}{1 - b/u} = \ln \frac{1}{1} = \ln 1 = 0.$$

Evaluate at the lower limit $u = s$:

$$\ln \frac{s-a}{s-b}.$$

Therefore:

$$\mathcal{L}\left\{\frac{e^{at} - e^{bt}}{t}\right\} = 0 - \ln \frac{s-a}{s-b} = \ln \frac{s-b}{s-a}.$$

What Did We Learn?

$\mathcal{L}\{(e^{at} - e^{bt})/t\} = \ln \dfrac{s-b}{s-a}$. The logarithm arises naturally from integrating partial fractions. The upper limit evaluates to $\ln 1 = 0$, leaving only the lower limit's contribution (with a sign flip).

7 Additional Applications

Further Results via Division by t

(a) $\mathcal{L}\{(1 - \cos t)/t\}$

Let $f(t) = 1 - \cos t$, so $F(s) = \mathcal{L}\{1\} - \mathcal{L}\{\cos t\} = \dfrac{1}{s} - \dfrac{s}{s^2+1}$.

Existence condition: $\displaystyle\lim_{t\to 0^+} \frac{1 - \cos t}{t} = \lim_{t\to 0} \frac{t^2/2 + O(t^4)}{t} = 0$ (finite). ✓

Apply the theorem:

$$\mathcal{L}\left\{\frac{1-\cos t}{t}\right\} = \int_s^\infty \left(\frac{1}{u} - \frac{u}{u^2+1}\right) du = \left[\ln u - \tfrac{1}{2}\ln(u^2+1)\right]_s^\infty.$$

Write $\ln u - \frac{1}{2}\ln(u^2+1) = \frac{1}{2}\ln\dfrac{u^2}{u^2+1}$. As $u \to \infty$: $\dfrac{u^2}{u^2+1} \to 1$, so the expression $\to \frac{1}{2}\ln 1 = 0$. At the lower limit $u = s$: $\frac{1}{2}\ln\dfrac{s^2}{s^2+1}$. Therefore:

$$\mathcal{L}\left\{\frac{1-\cos t}{t}\right\} = 0 - \frac{1}{2}\ln\frac{s^2}{s^2+1} = \frac{1}{2}\ln\frac{s^2+1}{s^2}.$$

(b) $\mathcal{L}\{\sin^2(t)/t^2\}$ (applying the theorem twice)
Step 1. Let $g(t) = \sin^2 t$. Using $\sin^2 t = (1-\cos 2t)/2$:

$$G(s) = \frac{1}{2}\left(\frac{1}{s} - \frac{s}{s^2+4}\right).$$

Check condition for first division by t: $\lim_{t\to 0}\sin^2(t)/t = 0$ (finite). ✓

$$\mathcal{L}\left\{\frac{\sin^2 t}{t}\right\} = \int_s^\infty G(u)\,du = \frac{1}{2}\int_s^\infty\left(\frac{1}{u} - \frac{u}{u^2+4}\right)du = \frac{1}{2}\Big[\ln u - \tfrac{1}{2}\ln(u^2+4)\Big]_s^\infty = \frac{1}{2}\cdot\frac{1}{2}\ln\frac{s^2+4}{s^2} = \frac{1}{4}\ln$$

Step 2. Let $h(t) = \sin^2(t)/t$; we just found $H(s) = \frac{1}{4}\ln\frac{s^2+4}{s^2}$. Check condition for second division by t: $\lim_{t\to 0^+}\dfrac{\sin^2 t}{t^2} = 1$ (finite). ✓

$$\mathcal{L}\left\{\frac{\sin^2 t}{t^2}\right\} = \int_s^\infty H(u)\,du = \frac{1}{4}\int_s^\infty \ln\frac{u^2+4}{u^2}\,du.$$

Using integration by parts ($\int \ln(1+4/u^2)\,du = u\ln(1+4/u^2) + 4\arctan(u/2)$) and evaluating from s to ∞ gives:

$$\mathcal{L}\left\{\frac{\sin^2 t}{t^2}\right\} = \frac{\pi}{2} - \arctan\left(\frac{s}{2}\right) - \frac{s}{4}\ln\left(1+\frac{4}{s^2}\right) = \operatorname{arccot}\left(\frac{s}{2}\right) - \frac{s}{4}\ln\left(\frac{s^2+4}{s^2}\right) \quad (s>0).$$

What Did We Learn?

Division by t can be applied iteratively provided the existence condition is re-verified at each stage. The result for $(1-\cos t)/t$ is $\frac{1}{2}\ln\big((s^2+1)/s^2\big)$, and for $\sin^2(t)/t^2$ a combination of logarithm and arctangent appears.

8 Comparison Table: Topics 10 vs. 11

Duality Table — Operations in Time and s-Domains

Operation	Time Domain	Effect in s-Domain
Multiply by t^n	$t^n f(t)$	$(-1)^n \dfrac{d^nF}{ds^n}$
Divide by t	$f(t)/t$	$\int_s^\infty F(u)\,du$
Differentiate	$f'(t)$	$sF(s) - f(0)$
Integrate	$\int_0^t f(\tau)\,d\tau$	$F(s)/s$

Big Picture

The four duality pairs above form the complete first-order operational calculus of the Laplace transform. Memorise them as two symmetric pairs: differentiation/integration in t and multiplication/division by t.

9 Worked Examples

Example 1: $\mathcal{L}\left\{\frac{\cos(at)-\cos(bt)}{t}\right\}$

Given: Find $\mathcal{L}\left\{\dfrac{\cos(at)-\cos(bt)}{t}\right\}$ for constants $a \neq b$ and $s > 0$.

Step 1 — Existence condition.

$$\lim_{t\to 0^+} \frac{\cos(at)-\cos(bt)}{t}.$$

Both $\cos(at) \to 1$ and $\cos(bt) \to 1$ as $t \to 0$, so the numerator $\to 0$. Apply L'Hôpital's rule:

$$\lim_{t\to 0} \frac{-a\sin(at)+b\sin(bt)}{1} = -a\cdot 0 + b\cdot 0 = 0 \quad \text{(finite).} \quad \checkmark$$

Step 2 — Identify $F(s)$.

$$F(s) = \mathcal{L}\{\cos(at)\} - \mathcal{L}\{\cos(bt)\} = \frac{s}{s^2+a^2} - \frac{s}{s^2+b^2}.$$

Step 3 — Apply division by t theorem.

$$\mathcal{L}\left\{\frac{\cos(at)-\cos(bt)}{t}\right\} = \int_s^\infty \left(\frac{u}{u^2+a^2} - \frac{u}{u^2+b^2}\right) du.$$

Step 4 — Integrate each term.

$$\int_s^\infty \frac{u}{u^2+a^2}\,du = \left[\frac{1}{2}\ln(u^2+a^2)\right]_s^\infty.$$

As $u \to \infty$: $\frac{1}{2}\ln(u^2+a^2) - \frac{1}{2}\ln(u^2+b^2) = \frac{1}{2}\ln\dfrac{u^2+a^2}{u^2+b^2} \to \frac{1}{2}\ln 1 = 0$.

At the lower limit $u = s$: $\frac{1}{2}\ln(s^2+a^2) - \frac{1}{2}\ln(s^2+b^2) = \frac{1}{2}\ln\dfrac{s^2+a^2}{s^2+b^2}$.

Step 5 — Combine.

$$\mathcal{L}\left\{\frac{\cos(at)-\cos(bt)}{t}\right\} = 0 - \frac{1}{2}\ln\frac{s^2+a^2}{s^2+b^2} = \frac{1}{2}\ln\frac{s^2+b^2}{s^2+a^2}.$$

What Did We Learn?

$\mathcal{L}\{(\cos(at)-\cos(bt))/t\} = \frac{1}{2}\ln\dfrac{s^2+b^2}{s^2+a^2}$. The existence condition passes (numerator $\to 0$ faster than denominator). Integrating $u/(u^2+c^2)$ produces $\frac{1}{2}\ln(u^2+c^2)$, leading to a logarithmic result.

Example 2: $\mathcal{L}\left\{\frac{\sin(2t)}{t}\right\}$

Given: Find $\mathcal{L}\left\{\dfrac{\sin 2t}{t}\right\}$.

Step 1 — Existence condition. $\lim_{t\to 0^+}\dfrac{\sin 2t}{t} = \lim_{t\to 0}\dfrac{2t-(2t)^3/6+\cdots}{t} = 2$ (finite). ✓

Step 2 — Identify $F(s)$.

$$F(s) = \mathcal{L}\{\sin 2t\} = \frac{2}{s^2+4}.$$

Step 3 — Apply theorem.

$$\mathcal{L}\left\{\frac{\sin 2t}{t}\right\} = \int_s^\infty \frac{2}{u^2+4}\,du = 2\int_s^\infty \frac{du}{u^2+4}.$$

Step 4 — Evaluate the integral. Recall $\displaystyle\int \frac{du}{u^2+a^2} = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + C$. With $a = 2$:

$$2\left[\frac{1}{2}\arctan\left(\frac{u}{2}\right)\right]_s^\infty = \left[\arctan\left(\frac{u}{2}\right)\right]_s^\infty.$$

Step 5 — Limits. $\lim_{u\to\infty}\arctan(u/2) = \pi/2$; at $u = s$: $\arctan(s/2)$.

$$\boxed{\mathcal{L}\left\{\frac{\sin 2t}{t}\right\} = \frac{\pi}{2} - \arctan\left(\frac{s}{2}\right).}$$

What Did We Learn?

For $\mathcal{L}\{\sin(at)/t\}$ the general pattern is $\dfrac{\pi}{2} - \arctan\left(\dfrac{s}{a}\right)$. When $a = 1$ this reduces to the sinc-function result from Section 5.

Example 3: $\mathcal{L}\left\{\frac{e^{-t}-e^{-2t}}{t}\right\}$

Given: Find $\mathcal{L}\left\{\dfrac{e^{-t}-e^{-2t}}{t}\right\}$.

Step 1 — Existence condition. $\displaystyle\lim_{t\to 0^+}\frac{e^{-t}-e^{-2t}}{t} = \lim_{t\to 0}\frac{(1-t+\cdots)-(1-2t+\cdots)}{t} = \lim_{t\to 0}\frac{t+O(t^2)}{t} = 1$ (finite). ✓

Step 2 — Identify $F(s)$.

$$F(s) = \frac{1}{s+1} - \frac{1}{s+2}.$$

Step 3 — Apply theorem.

$$\mathcal{L}\left\{\frac{e^{-t}-e^{-2t}}{t}\right\} = \int_s^\infty \left(\frac{1}{u+1} - \frac{1}{u+2}\right) du = [\ln(u+1) - \ln(u+2)]_s^\infty = \left[\ln\frac{u+1}{u+2}\right]_s^\infty.$$

Step 4 — Evaluate limits.

$$\lim_{u\to\infty} \ln\frac{u+1}{u+2} = \ln 1 = 0.$$

At $u = s$: $\ln\dfrac{s+1}{s+2}$.

Step 5 — Combine and simplify.

$$= 0 - \ln\frac{s+1}{s+2} = \ln\frac{s+2}{s+1}.$$

This is the special case $a = -1$, $b = -2$ of the general formula $\ln\dfrac{s-b}{s-a}$.

What Did We Learn?

$\mathcal{L}\{(e^{-t}-e^{-2t})/t\} = \ln\dfrac{s+2}{s+1}$. Matching to the general result $\ln\dfrac{s-b}{s-a}$ with $a = -1$, $b = -2$ confirms the formula and serves as a useful self-check.

10 Excel Example (Numerical Verification of $\mathcal{L}\{\sin(t)/t\}$ at $s = 1$)

Objective. Numerically estimate $\displaystyle\int_0^\infty e^{-t}\frac{\sin t}{t}\,dt$ using the trapezoidal rule in Excel and compare to the exact value $\pi/2 - \arctan(1) = \pi/2 - \pi/4 = \pi/4 \approx 0.7854$.

Spreadsheet Layout.

A t	**B** $\sin(t)/t$	**C** $e^{-t}\sin(t)/t$	**D** Trap. weight $h/2$	**E** Cumulative sum
0.001	`=SIN(A2)/A2`	`=EXP(-A2)*B2`	`=0.1/2`	0
0.101	`=SIN(A3)/A3`	`=EXP(-A3)*B3`	`=0.1/2`	`=E2+D2*(C2+C3)`
0.201	`=SIN(A4)/A4`	`=EXP(-A4)*B4`	`=0.1/2`	`=E3+D3*(C3+C4)`
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
20.001	`=SIN(A202)/A202`	`=EXP(-A202)*B202`	`=0.1/2`	(final sum)

Step-by-step instructions.

1. In **A2** enter `0.001` (avoid $t = 0$ to prevent division by zero). In **A3** enter `=A2+0.1`. Fill down to $t \approx 20$.
2. **Column B**: `=SIN(A2)/A2`. Fill down.
3. **Column C**: `=EXP(-A2)*B2` (this is the integrand $e^{-st}\sin(t)/t$ at $s = 1$). Fill down.
4. **Column D**: enter the trapezoid half-step weight `=0.1/2` in each row.
5. **Column E** (**E2**): `0` (initial cumulative integral at t_0).
   E3 onwards: `=E2+D2*(C2+C3)`. This accumulates the trapezoidal approximation.
6. Read the last entry in column E: it should be close to 0.7854.
7. In a nearby cell enter the exact value: `=PI()/4` (≈ 0.7854).
8. Compute the relative error: `=(E_final-PI()/4)/(PI()/4)*100%`.

Sample rows (representative values):

t	$\sin(t)/t$	$e^{-t}\sin(t)/t$	$h/2$	Cum. Sum
0.001	0.99999983	0.99899983	0.05	0.00000
0.101	0.99829983	0.90238120	0.05	0.09985
1.001	0.84114701	0.30932822	0.05	0.58432
10.001	−0.054399	−0.0000025	0.05	0.78538
20.001	0.045647	0.0000000	0.05	0.78539

What Did We Learn?

The trapezoidal sum converges to ≈ 0.7854, confirming $\mathcal{L}\{\sin t/t\}\big|_{s=1} = \pi/4$. Numerical verification builds confidence in symbolic results and reveals how rapidly the integrand decays once e^{-t} is included.

11 Python Example

Listing 1: Division by t — Symbolic and Numerical

```python
import sympy as sp
import numpy as np
import matplotlib.pyplot as plt

# -- 1. Symbolic computation of L{sin(t)/t} ----------------------------------
u, s = sp.symbols('u s', positive=True)

F_u = 1 / (u**2 + 1)                          # F(u) = L{sin t} at u
transform_expr = sp.integrate(F_u, (u, s, sp.oo))
transform_simplified = sp.simplify(transform_expr)
print("Symbolic result: L{sin(t)/t} =", transform_simplified)
# Expected output: Symbolic result: L{sin(t)/t} = pi/2 - atan(s)

# -- 2. Numerical evaluation at selected s values ----------------------------
s_vals = [0.0, 0.5, 1.0, 2.0, 5.0]
exact_formula = lambda sv: np.pi/2 - np.arctan(sv)

print("\ns      Exact (pi/2 - arctan(s))   SymPy numeric")
print("-" * 50)
for sv in s_vals:
    exact = exact_formula(sv)
    sympy_val = float(transform_simplified.subs(s, sv)) if sv > 0 else float(np.pi/2)
    print(f"{sv:.1f}    {exact:.6f}                  {sympy_val:.6f}")
# Expected output:
# s      Exact (pi/2 - arctan(s))   SymPy numeric
# --------------------------------------------------
# 0.0    1.570796                  1.570796
# 0.5    1.107149                  1.107149
# 1.0    0.785398                  0.785398
# 2.0    0.463648                  0.463648
# 5.0    0.197396                  0.197396

# -- 3. Plot the transform function pi/2 - arctan(s) -------------------------
s_plot = np.linspace(0.01, 10, 400)
F_plot = np.pi/2 - np.arctan(s_plot)

plt.figure(figsize=(8, 4))
plt.plot(s_plot, F_plot, 'b-', linewidth=2,
         label=r'$\mathcal{L}\{\sin(t)/t\} = \pi/2 - \arctan(s)$')
plt.axhline(y=np.pi/2, color='gray', linestyle='--', alpha=0.5,
            label=r'$\pi/2$ (asymptote as $s\to 0^+$)')
plt.axhline(y=0, color='black', linewidth=0.5)
plt.xlabel('s')
```

```python
plt.ylabel(r'$F(s)$')
plt.title(r'Laplace Transform of $\sin(t)/t$')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig('sinc_transform.png', dpi=150)
plt.show()
print("Plot saved as sinc_transform.png")
# Expected: a decreasing curve from pi/2 at s=0 towards 0 as s ->
    inf
```

What Did We Learn?

SymPy confirms $\mathcal{L}\{\sin(t)/t\} = \pi/2 - \arctan(s)$ symbolically. The numerical table shows the transform is a strictly decreasing function of s, starting at $\pi/2$ when $s = 0$ and approaching 0 as $s \to \infty$. The matplotlib plot visualises this decay.

12 Viva-Style Oral Questions

Q1. State the division-by-t theorem.
If $\mathcal{L}\{f(t)\} = F(s)$ and $\lim_{t\to 0^+} f(t)/t$ is finite, then $\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u)\,du$.

Q2. Give a sketch proof of the division-by-t theorem.
Write $\int_s^\infty F(u)\,du = \int_s^\infty \int_0^\infty e^{-ut} f(t)\,dt\,du$. Because both integrals converge absolutely (Fubini), swap the order to get $\int_0^\infty f(t)\left[\int_s^\infty e^{-ut}\,du\right] dt$. The inner integral equals e^{-st}/t, yielding $\int_0^\infty e^{-st} f(t)/t\,dt = \mathcal{L}\{f(t)/t\}$.

Q3. What is the existence condition and why is it needed?
The limit $\lim_{t\to 0^+} f(t)/t$ must be finite (equivalently $f(0) = 0$). If $f(0) \neq 0$, the integrand $e^{-st} f(t)/t \sim f(0)e^{-st}/t$ is not integrable near $t = 0$, so the Laplace integral diverges.

Q4. Derive $\mathcal{L}\{\sin t/t\}$.
$F(s) = 1/(s^2 + 1)$; existence condition: $\lim_{t\to 0} \sin(t)/t = 1$ (OK). $\int_s^\infty du/(u^2 + 1) = [\arctan u]_s^\infty = \pi/2 - \arctan(s)$.

Q5. Why does the integration run from s to ∞ and not from 0 to s?
The proof shows $\int_s^\infty F(u)\,du = \mathcal{L}\{f(t)/t\}$; the lower limit is the current value of s because that is where the inner integral over u starts in the Fubini argument. Integrating from 0 to s would give a completely different — and incorrect — quantity.

Q6. How does this formula differ from Topic 09 (Laplace of integrals)?
Topic 09 states $\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = F(s)/s$. Division by t (Topic 11) gives $\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u)\,du$. One divides in s; the other integrates in s from s to ∞. They are entirely different operations.

Q7. What is the sinc function, and where does it appear?
The (unnormalised) sinc function is $\mathrm{sinc}(t) = \sin(t)/t$ (defined as 1 at $t = 0$ by the limit). It is the impulse response of an ideal low-pass filter in communication and

signal processing, and appears in the Dirichlet integral $\int_0^\infty \sin(t)/t\,dt = \pi/2$. Its Laplace transform is $\pi/2 - \arctan(s)$.

Q8. State the result for $\mathcal{L}\{(e^{at} - e^{bt})/t\}$.
Provided $s > \max(a, b)$: $\mathcal{L}\big\{(e^{at} - e^{bt})/t\big\} = \ln \dfrac{s-b}{s-a}$. The logarithm arises from integrating the difference of simple fractions $1/(u-a) - 1/(u-b)$, and the upper limit evaluates to $\ln 1 = 0$.

13 Descriptive Questions

D1. State and prove the division-by-t theorem in detail.
State: $\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u)\,du$ given $\lim_{t\to 0^+} f(t)/t$ finite. Proof: Expand $\int_s^\infty F(u)\,du$ using the Laplace definition, invoke Fubini's theorem to interchange the order of integration, evaluate the inner integral $\int_s^\infty e^{-ut}\,du = e^{-st}/t$, and read off the Laplace definition of $f(t)/t$.

D2. Find $\mathcal{L}\{\sin(at)/t\}$ for $a > 0$, showing all steps.
$F(s) = a/(s^2 + a^2)$. Existence: $\lim_{t\to 0} \sin(at)/t = a$ (finite). ✓

$$\mathcal{L}\left\{\frac{\sin(at)}{t}\right\} = \int_s^\infty \frac{a}{u^2 + a^2}\,du = a \cdot \frac{1}{a}\Big[\arctan(u/a)\Big]_s^\infty = \frac{\pi}{2} - \arctan\left(\frac{s}{a}\right).$$

D3. Find $\mathcal{L}\{(e^{2t} - e^{3t})/t\}$, showing all steps.
$F(s) = 1/(s-2) - 1/(s-3)$ (valid for $s > 3$). Existence: $\lim_{t\to 0}(e^{2t} - e^{3t})/t = 2 - 3 = -1$ (finite). ✓

$$\int_s^\infty [1/(u-2) - 1/(u-3)]\,du = [\ln(u-2) - \ln(u-3)]_s^\infty = 0 - \ln\frac{s-2}{s-3} = \ln\frac{s-3}{s-2}.$$

D4. Explain the existence condition with a concrete counterexample.
The existence condition requires $\lim_{t\to 0^+} f(t)/t$ to be finite. Counterexample: $f(t) = \cos t$, so $f(0) = 1 \neq 0$. Then $\cos(t)/t \sim 1/t \to \infty$ as $t \to 0^+$. The integral $\int_0^\infty e^{-st}\cos(t)/t\,dt$ diverges (the integrand is not Lebesgue-integrable near $t = 0$), so $\mathcal{L}\{\cos(t)/t\}$ does not exist. In contrast, $f(t) = \sin t$ satisfies $\sin(0) = 0$, so the condition holds.

D5. State the complete duality table of Topics 08–11 and explain the symmetry.
The four fundamental operations form two dual pairs:

Time domain	s-domain	Topic
$f^{(n)}(t)$ (differentiate)	$s^n F(s) - \cdots$	08
$\int_0^t f\,d\tau$ (integrate)	$F(s)/s$	09
$t^n f(t)$ (multiply by t^n)	$(-1)^n d^n F/ds^n$	10
$f(t)/t$ (divide by t)	$\int_s^\infty F(u)\,du$	11

The symmetry: differentiation/integration in $t \leftrightarrow$ algebraic ops in s; multiplication/division by $t \leftrightarrow$ differentiation/integration in s. This duality makes the Laplace transform a "dictionary" between two domains, each having its own analogue of the fundamental theorem of calculus.

14 Practice Problems

P1. Find $\mathcal{L}\left\{\dfrac{1-e^{-t}}{t}\right\}$.

Hint: $F(s) = 1/s - 1/(s+1)$; existence: $\lim_{t\to 0}(1-e^{-t})/t = 1$. Integrate: $\ln(s+1) - \ln s + C$; evaluate limits to get $\ln \dfrac{s+1}{s}$.

P2. Find $\mathcal{L}\left\{\dfrac{\sin t \cos t}{t}\right\}$ (use double-angle formula first).

Hint: $\sin t \cos t = \frac{1}{2}\sin 2t$, so $\mathcal{L}\left\{\dfrac{\sin t\cos t}{t}\right\} = \frac{1}{2}\mathcal{L}\left\{\dfrac{\sin 2t}{t}\right\} = \frac{1}{2}\left(\dfrac{\pi}{2} - \arctan\left(\dfrac{s}{2}\right)\right)$.

P3. Find $\mathcal{L}\left\{\dfrac{e^{-at}-e^{-bt}}{t}\right\}$ for $a, b > 0$ and $s > 0$.

Hint: $\mathcal{L}\{e^{-at} - e^{-bt}\} = \dfrac{1}{s+a} - \dfrac{1}{s+b}$; integrating from s to ∞ gives $\ln\left(\dfrac{s+b}{s+a}\right)$.

P4. Does $\mathcal{L}\left\{\dfrac{\cos(at)}{t}\right\}$ exist? Justify fully.

Hint: Check the existence condition: $\lim_{t\to 0^+} \dfrac{\cos(at)}{t} = \dfrac{1}{0^+} = \infty$. The condition *fails*; the transform does *not* exist.

P5. Find $\mathcal{L}\left\{\dfrac{\sin^2 t}{t}\right\}$.

Hint: Use $\sin^2 t = (1-\cos 2t)/2$, so $\mathcal{L}\{\sin^2 t\} = \frac{1}{2}(1/s - s/(s^2+4))$. Integrate from s to ∞: result is $\dfrac{1}{4}\ln\dfrac{s^2+4}{s^2}$.

P6. Find $\mathcal{L}\left\{\dfrac{\cosh(at)-1}{t}\right\}$.

Hint: $\mathcal{L}\{\cosh(at)-1\} = \dfrac{s}{s^2-a^2} - \dfrac{1}{s}$. Existence: $(\cosh(at)-1)/t \to 0$ as $t \to 0$ (since $\cosh(0) = 1$). Integrate from s to ∞: $\dfrac{1}{2}\ln\left(\dfrac{s^2}{s^2-a^2}\right)$ (for $s > a > 0$).

15 MCQs

MCQ1. The division-by-t formula states that $\mathcal{L}\{f(t)/t\}$ equals:

(a) $\displaystyle\int_0^s F(u)\,du$

(b) $\displaystyle\int_0^\infty F(u)\,du$

(c) $\displaystyle\int_s^\infty F(u)\,du$

(d) $F(s)/s$

Explanation: The integral runs from s to ∞; option (a) has the wrong direction and wrong lower limit; (d) is Topic 09's formula.

MCQ2. The existence condition for $\mathcal{L}\{f(t)/t\}$ to be valid is:

(a) $f(\infty) = 0$

(b) $\lim_{t\to 0^+} f(t)/t$ **is finite**

(c) $f(t)$ is differentiable for all $t > 0$

(d) $F(s)$ is bounded

Explanation: If $f(0) \neq 0$, the integrand $e^{-st} f(t)/t$ diverges near $t = 0$, making the Laplace integral undefined.

MCQ3. $\mathcal{L}\left\{\dfrac{\sin t}{t}\right\}$ equals:

(a) $\arctan(s)$

(b) $\ln(s^2 + 1)$

(c) $\dfrac{1}{s^2 + 1}$

(d) $\dfrac{\pi}{2} - \arctan(s)$

Explanation: Integrating $1/(u^2+1)$ from s to ∞ gives $[\arctan u]_s^\infty = \pi/2 - \arctan(s)$.

MCQ4. Compared with the multiplication-by-t formula (Topic 10), dividing by t corresponds to which s-domain operation?

(a) Differentiating $F(s)$

(b) Multiplying $F(s)$ by s

(c) Dividing $F(s)$ by s

(d) **Integrating $F(s)$ from s to ∞**

Explanation: Multiply by $t \leftrightarrow$ differentiate; divide by $t \leftrightarrow$ integrate from s to ∞. These are dual operations.

MCQ5. If $f(0) \neq 0$, applying the division-by-t formula:

(a) Gives the correct result if s is large enough

(b) Gives a result that is off by a constant

(c) **Fails — the Laplace transform of $f(t)/t$ does not exist**

(d) Requires a correction term $f(0)/s$

Explanation: When $f(0) \neq 0$, $f(t)/t \sim f(0)/t$ near $t = 0$, which is not Laplace-transformable; the integral diverges regardless of s.

16 Common Mistakes

Common Mistakes in Division by t

Mistake	What Students Do	Correct Approach
Wrong integration direction	Write $\int_0^s F(u)\,du$ instead of $\int_s^\infty F(u)\,du$	Always integrate from the current value s upward to ∞; the inner integral in the Fubini proof runs from s to ∞
Skipping the existence check	Apply $\mathcal{L}\{f(t)/t\}$ without verifying $\lim_{t\to 0^+} f(t)/t$ is finite	Always check the limit first; if $f(0) \neq 0$ the formula cannot be used
Confusing with Topic 09	Write $F(s)/s$ (Topic 09 result) when the formula requires $\int_s^\infty F(u)\,du$	Topic 09: $\mathcal{L}\{\int_0^t f\} = F(s)/s$; Topic 11: $\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u)\,du$ — completely different
Forgetting the upper limit	Write $[\text{antiderivative}]_s$ without evaluating $\lim_{u\to\infty}$	Always evaluate both limits; the upper limit often simplifies to 0 (e.g. $\ln 1 = 0$, $\arctan(\infty) = \pi/2$)
Attempting $\mathcal{L}\{\cos(t)/t\}$	Apply the formula assuming it always works	Check: $\lim_{t\to 0} \cos(t)/t = \infty$ — existence condition fails; $\mathcal{L}\{\cos(t)/t\}$ does not exist

17 Quick Recap

Quick Recap — Topic 11: Division by t

- **Division by t formula:** $\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u)\,du$ (integration from s to ∞ in the s-domain).
- **Existence condition:** $\lim_{t\to 0^+} f(t)/t$ must be finite; equivalently $f(0) = 0$. Fails for $\cos(t)/t$, $1/t$, etc.
- **Key result — sinc function:** $\mathcal{L}\{\sin(t)/t\} = \pi/2 - \arctan(s)$; evaluate via $\int_s^\infty du/(u^2+1)$.
- **Key result — exponential ratio:** $\mathcal{L}\big\{(e^{at} - e^{bt})/t\big\} = \ln\big[(s-b)/(s-a)\big]$; arises from integrating $1/(u-a) - 1/(u-b)$.
- **Complete duality table (Topics 08–11):** differentiate $f \leftrightarrow$ multiply $F(s)$ by s; integrate $f \leftrightarrow$ divide $F(s)$ by s; multiply f by $t^n \leftrightarrow$ differentiate $F(s)$ n times; divide f by $t \leftrightarrow$ integrate $F(s)$ from s to ∞.
- **Integration direction is s to ∞**, not 0 to s — the Fubini argument fixes this direction.
- **Upper limit evaluation:** $\ln(u-a)/(u-b) \to \ln 1 = 0$ and $\arctan(\infty) = \pi/2$ are the two standard upper-limit values you will need.

- **Link to Topic 12:** Setting $s = 0$ in $\mathcal{L}\{f(t)/t\} = \int_0^\infty F(u)\,du$ evaluates definite integrals like $\int_0^\infty \sin(t)/t\,dt = \pi/2$ without contour integration.